

Final datasheet

PrimePACK™3+ B-series module with TRENCHSTOP™ IGBT7 and emitter controlled 7 diode and NTC

Features

- Electrical features
 - $V_{CES} = 2300\text{ V}$
 - $I_{C\text{ nom}} = 1800\text{ A} / I_{CRM} = 3600\text{ A}$
 - TRENCHSTOP™ IGBT7
 - $T_{vj,op} = 150^{\circ}\text{C}$
 - Overload operation up to 175°C
 - Low $V_{CE,sat}$
 - Low switching losses
 - High current density
 - Low inductive design
- Mechanical features
 - Package with CTI > 400
 - High creepage and clearance distances
 - High power density



Potential applications

- Three-level applications
- Solar applications

Product validation

- Qualified for industrial applications according to the relevant tests of IEC 60747, 60749 and 60068

Description

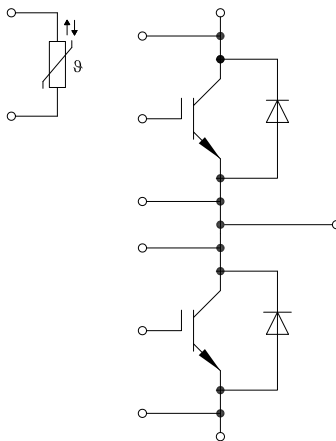




Table of contents

	Table of contents	2
1	Package	3
2	IGBT, Inverter	3
3	Diode, Inverter	5
4	IGBT, 3-Level	6
5	Diode, 3-Level	7
6	NTC-Thermistor	8
7	Characteristics diagrams	10
8	Circuit diagram	16
9	Package outlines	17
10	Module label code	18
	Revision history	19
	Disclaimer	20

1 Package

Table 1 Insulation coordination

Parameter	Symbol	Note or test condition	Values	Unit
Isolation test voltage	V_{ISOL}	RMS, $f = 50$ Hz	4.0	kV
Material of module baseplate			Cu	
Comparative tracking index	CTI		> 400	

Table 2 Characteristic values

Parameter	Symbol	Note or test condition		Values			Unit
				Min.	Typ.	Max.	
Stray inductance module	L_{sCE}				10		nH
Module lead resistance, terminals - chip	$R_{AA'+CC'}$	$T_C = 25\text{ }^{\circ}\text{C}$, per switch			0.09		mΩ
Module lead resistance, terminals - chip	$R_{CC'+EE'}$	$T_C = 25\text{ }^{\circ}\text{C}$, per switch			0.1		mΩ
Storage temperature	T_{stg}			-40		150	°C
Maximum baseplate operation temperature	T_{BPmax}					150	°C
Mounting torque for module mounting	M	- Mounting according to valid application note	M5, Screw	3		6	Nm
Terminal connection torque	M	- Mounting according to valid application note	M4, Screw	1.8		2.1	Nm
			M8, Screw	8		10	
Weight	G				1400		g

2 IGBT, Inverter

Table 3 Maximum rated values

Parameter	Symbol	Note or test condition		Values	Unit
Collector-emitter voltage	V_{CES}		$T_{vj} = 25$ °C	2300	V
Implemented collector current	I_{CN}			1800	A
Continuous DC collector current	I_{CDC}	$T_{vj max} = 150$ °C	$T_C = 50$ °C	1800	A
Repetitive peak collector current	I_{CRM}	t_p limited by $T_{vj op}$		3600	A
Gate-emitter peak voltage	V_{GES}			±20	V

Table 4 **Characteristic values**

Parameter	Symbol	Note or test condition		Values			Unit
				Min.	Typ.	Max.	
Collector-emitter saturation voltage	$V_{CE\ sat}$	$I_C = 1800\ A, V_{GE} = 15\ V$	$T_{vj} = 25\ ^\circ C$		1.80	2.26	V
			$T_{vj} = 125\ ^\circ C$		2.15	2.94	
			$T_{vj} = 150\ ^\circ C$		2.25	3.13	
Gate threshold voltage	V_{GETh}	$I_C = 49.5\ mA, V_{CE} = V_{GE}, T_{vj} = 25\ ^\circ C$		5.15	5.80	6.45	V
Gate charge	Q_G	$V_{GE} = \pm 15\ V, V_{CC} = 1200\ V, T_{vj} = 25\ ^\circ C$			14.6		μC
Internal gate resistor	R_{Gint}	$T_{vj} = 25\ ^\circ C$			0.96		Ω
Input capacitance	C_{ies}	$f = 100\ kHz, T_{vj} = 25\ ^\circ C, V_{CE} = 25\ V, V_{GE} = 0\ V$			225		nF
Reverse transfer capacitance	C_{res}	$f = 100\ kHz, T_{vj} = 25\ ^\circ C, V_{CE} = 25\ V, V_{GE} = 0\ V$			0.54		nF
Collector-emitter cut-off current	I_{CES}	$V_{CE} = 2300\ V, V_{GE} = 0\ V$	$T_{vj} = 125\ ^\circ C$			30	mA
Gate-emitter leakage current	I_{GES}	$V_{CE} = 0\ V, V_{GE} = 20\ V, T_{vj} = 25\ ^\circ C$				400	nA
Turn-on delay time (inductive load)	t_{don}	$I_C = 1800\ A, V_{CC} = 1200\ V, V_{GE} = \pm 15\ V, R_{Gon} = 0.1\ \Omega$	$T_{vj} = 25\ ^\circ C$		0.53		μs
			$T_{vj} = 125\ ^\circ C$		0.55		
			$T_{vj} = 150\ ^\circ C$		0.56		
Rise time (inductive load)	t_r	$I_C = 1800\ A, V_{CC} = 1200\ V, V_{GE} = \pm 15\ V, R_{Gon} = 0.1\ \Omega$	$T_{vj} = 25\ ^\circ C$		0.072		μs
			$T_{vj} = 125\ ^\circ C$		0.078		
			$T_{vj} = 150\ ^\circ C$		0.083		
Turn-off delay time (inductive load)	t_{doff}	$I_C = 1800\ A, V_{CC} = 1200\ V, V_{GE} = \pm 15\ V, R_{Goff} = 1.5\ \Omega$	$T_{vj} = 25\ ^\circ C$		0.955		μs
			$T_{vj} = 125\ ^\circ C$		1.05		
			$T_{vj} = 150\ ^\circ C$		1.08		
Fall time (inductive load)	t_f	$I_C = 1800\ A, V_{CC} = 1200\ V, V_{GE} = \pm 15\ V, R_{Goff} = 1.5\ \Omega$	$T_{vj} = 25\ ^\circ C$		0.77		μs
			$T_{vj} = 125\ ^\circ C$		1.02		
			$T_{vj} = 150\ ^\circ C$		1.1		
Turn-on time (resistive load)	t_{on_R}	$I_C = 500\ A, V_{CC} = 2000\ V, V_{GE} = \pm 15\ V, R_{Gon} = 0.1\ \Omega$	$T_{vj} = 25\ ^\circ C$	0.79			μs
Turn-on energy loss per pulse	E_{on}	$I_C = 1800\ A, V_{CC} = 1200\ V, L_\sigma = 20\ nH, V_{GE} = \pm 15\ V, R_{Gon} = 0.1\ \Omega, di/dt = 17500\ A/\mu s (T_{vj} = 150\ ^\circ C)$	$T_{vj} = 25\ ^\circ C$		570		mJ
			$T_{vj} = 125\ ^\circ C$		815		
			$T_{vj} = 150\ ^\circ C$		915		
Turn-off energy loss per pulse	E_{off}	$I_C = 1800\ A, V_{CC} = 1200\ V, L_\sigma = 20\ nH, V_{GE} = \pm 15\ V, R_{Goff} = 1.5\ \Omega, dv/dt = 4050\ V/\mu s (T_{vj} = 150\ ^\circ C)$	$T_{vj} = 25\ ^\circ C$		885		mJ
			$T_{vj} = 125\ ^\circ C$		1160		
			$T_{vj} = 150\ ^\circ C$		1240		

(table continues...)

Table 4 (continued) Characteristic values

Parameter	Symbol	Note or test condition	Values			Unit
			Min.	Typ.	Max.	
SC data	I_{SC}	$V_{GE} \leq 15 \text{ V}$, $V_{CC} = 1200 \text{ V}$, $V_{CEmax} = V_{CES} - L_{sCE} \cdot di/dt$ $t_p \leq 7 \mu\text{s}$, $T_{vj} = 150 \text{ }^\circ\text{C}$		8000		A
Thermal resistance, junction to case	R_{thJC}	per IGBT			17.7	K/kW
Thermal resistance, case to heat sink	R_{thCH}	per IGBT, $\lambda_{grease} = 1 \text{ W/(m}\cdot\text{K)}$		10.4		K/kW
Temperature under switching conditions	$T_{vj op}$		-40		150	$^\circ\text{C}$

3 Diode, Inverter

Table 5 Maximum rated values

Parameter	Symbol	Note or test condition	Values	Unit
Repetitive peak reverse voltage	V_{RRM}	$T_{vj} = 25 \text{ }^\circ\text{C}$	2300	V
Continuous DC forward current	I_F		1800	A
Repetitive peak forward current	I_{FRM}	$t_p = 1 \text{ ms}$	3600	A
I^2t - value	I^2t	$t_p = 10 \text{ ms}$, $V_R = 0 \text{ V}$	$T_{vj} = 125 \text{ }^\circ\text{C}$	kA^2s
			$T_{vj} = 150 \text{ }^\circ\text{C}$	
Maximum power dissipation	P_{RQM}	$T_{vj} = 150 \text{ }^\circ\text{C}$	2700	kW

Table 6 Characteristic values

Parameter	Symbol	Note or test condition	Values			Unit
			Min.	Typ.	Max.	
Forward voltage	V_F	$I_F = 1800 \text{ A}$, $V_{GE} = 0 \text{ V}$	$T_{vj} = 25 \text{ }^\circ\text{C}$	3.25	3.64	V
			$T_{vj} = 125 \text{ }^\circ\text{C}$	3.00	3.33	
			$T_{vj} = 150 \text{ }^\circ\text{C}$	2.95	3.22	
Peak reverse recovery current	I_{RM}	$V_{CC} = 1200 \text{ V}$, $I_F = 1800 \text{ A}$, $V_{GE} = -15 \text{ V}$, $-di_F/dt = 17500 \text{ A}/\mu\text{s}$ ($T_{vj} = 150 \text{ }^\circ\text{C}$)	$T_{vj} = 25 \text{ }^\circ\text{C}$	1700		A
			$T_{vj} = 125 \text{ }^\circ\text{C}$	1870		
			$T_{vj} = 150 \text{ }^\circ\text{C}$	1880		
Recovered charge	Q_r	$V_{CC} = 1200 \text{ V}$, $I_F = 1800 \text{ A}$, $V_{GE} = -15 \text{ V}$, $-di_F/dt = 17500 \text{ A}/\mu\text{s}$ ($T_{vj} = 150 \text{ }^\circ\text{C}$)	$T_{vj} = 25 \text{ }^\circ\text{C}$	300		μC
			$T_{vj} = 125 \text{ }^\circ\text{C}$	625		
			$T_{vj} = 150 \text{ }^\circ\text{C}$	740		

(table continues...)

Table 6 (continued) Characteristic values

Parameter	Symbol	Note or test condition	Values			Unit
			Min.	Typ.	Max.	
Reverse recovery energy	E_{rec}	$V_{CC} = 1200\text{ V}$, $I_F = 1800\text{ A}$, $V_{GE} = -15\text{ V}$, $-di_F/dt = 17500\text{ A}/\mu\text{s}$ ($T_{vj} = 150\text{ °C}$)	$T_{vj} = 25\text{ °C}$	240		mJ
			$T_{vj} = 125\text{ °C}$	500		
			$T_{vj} = 150\text{ °C}$	590		
Thermal resistance, junction to case	R_{thJC}	per diode			39.3	K/kW
Thermal resistance, case to heat sink	R_{thCH}	per diode, $\lambda_{grease} = 1\text{ W}/(\text{m}\cdot\text{K})$		15.4		K/kW
Temperature under switching conditions	$T_{vj\text{ op}}$		-40		150	°C

4 IGBT, 3-Level

Table 7 Maximum rated values

Parameter	Symbol	Note or test condition	Values	Unit
Collector-emitter voltage	V_{CES}	$T_{vj} = 25\text{ °C}$	2300	V
Continuous DC collector current	I_{CDC}	$T_C = 50\text{ °C}$	1800	A
Repetitive peak collector current	I_{CRM}	t_p limited by $T_{vj\text{ op}}$	3600	A
Gate-emitter peak voltage	V_{GES}		±20	V

Table 8 Characteristic values

Parameter	Symbol	Note or test condition	Values			Unit
			Min.	Typ.	Max.	
Collector-emitter saturation voltage	$V_{CE\text{ sat}}$	$I_C = 1800\text{ A}$, $V_{GE} = 15\text{ V}$	$T_{vj} = 25\text{ °C}$	1.80	2.26	V
			$T_{vj} = 125\text{ °C}$	2.15	2.94	
			$T_{vj} = 150\text{ °C}$	2.25	3.13	
Gate threshold voltage	V_{GEth}	$I_C = 49.5\text{ mA}$, $V_{CE} = V_{GE}$, $T_{vj} = 25\text{ °C}$	5.15	5.80	6.45	V
Gate charge	Q_G	$V_{GE} = \pm 15\text{ V}$, $V_{CC} = 1200\text{ V}$, $T_{vj} = 25\text{ °C}$		14.6		μC
Internal gate resistor	R_{Gint}	$T_{vj} = 25\text{ °C}$		0.96		Ω
Input capacitance	C_{ies}	$f = 100\text{ kHz}$, $T_{vj} = 25\text{ °C}$, $V_{CE} = 25\text{ V}$, $V_{GE} = 0\text{ V}$		225		nF
Reverse transfer capacitance	C_{res}	$f = 100\text{ kHz}$, $T_{vj} = 25\text{ °C}$, $V_{CE} = 25\text{ V}$, $V_{GE} = 0\text{ V}$		0.54		nF
Collector-emitter cut-off current	I_{CES}	$V_{CE} = 2300\text{ V}$, $V_{GE} = 0\text{ V}$	$T_{vj} = 125\text{ °C}$		30	mA

(table continues...)

Table 8 (continued) **Characteristic values**

Parameter	Symbol	Note or test condition	Values			Unit
			Min.	Typ.	Max.	
Gate-emitter leakage current	I_{GES}	$V_{CE} = 0 \text{ V}$, $V_{GE} = 20 \text{ V}$, $T_{vj} = 25 \text{ °C}$			400	nA
Turn-on delay time (inductive load)	t_{don}	$I_C = 1800 \text{ A}$, $V_{CC} = 600 \text{ V}$, $V_{GE} = \pm 15 \text{ V}$, $R_{Gon} = 0.1 \text{ } \Omega$	$T_{vj} = 25 \text{ °C}$	0.555		μs
			$T_{vj} = 125 \text{ °C}$	0.58		
			$T_{vj} = 150 \text{ °C}$	0.59		
Rise time (inductive load)	t_r	$I_C = 1800 \text{ A}$, $V_{CC} = 600 \text{ V}$, $V_{GE} = \pm 15 \text{ V}$, $R_{Gon} = 0.1 \text{ } \Omega$	$T_{vj} = 25 \text{ °C}$	0.19		μs
			$T_{vj} = 125 \text{ °C}$	0.205		
			$T_{vj} = 150 \text{ °C}$	0.215		
Turn-off delay time (inductive load)	t_{doff}	$I_C = 1800 \text{ A}$, $V_{CC} = 600 \text{ V}$, $V_{GE} = \pm 15 \text{ V}$, $R_{Goff} = 1.5 \text{ } \Omega$	$T_{vj} = 25 \text{ °C}$	0.885		μs
			$T_{vj} = 125 \text{ °C}$	0.955		
			$T_{vj} = 150 \text{ °C}$	0.98		
Fall time (inductive load)	t_f	$I_C = 1800 \text{ A}$, $V_{CC} = 600 \text{ V}$, $V_{GE} = \pm 15 \text{ V}$, $R_{Goff} = 1.5 \text{ } \Omega$	$T_{vj} = 25 \text{ °C}$	0.63		μs
			$T_{vj} = 125 \text{ °C}$	0.875		
			$T_{vj} = 150 \text{ °C}$	0.93		
Turn-on energy loss per pulse	E_{on}	$I_C = 1800 \text{ A}$, $V_{CC} = 600 \text{ V}$, $L_\sigma = 50 \text{ nH}$, $V_{GE} = \pm 15 \text{ V}$, $R_{Gon} = 0.1 \text{ } \Omega$, $di/dt = 6700 \text{ A}/\mu\text{s}$ ($T_{vj} = 150 \text{ °C}$)	$T_{vj} = 25 \text{ °C}$	240		mJ
			$T_{vj} = 125 \text{ °C}$	380		
			$T_{vj} = 150 \text{ °C}$	430		
Turn-off energy loss per pulse	E_{off}	$I_C = 1800 \text{ A}$, $V_{CC} = 600 \text{ V}$, $L_\sigma = 50 \text{ nH}$, $V_{GE} = \pm 15 \text{ V}$, $R_{Goff} = 1.5 \text{ } \Omega$, $dv/dt = 3150 \text{ V}/\mu\text{s}$ ($T_{vj} = 150 \text{ °C}$)	$T_{vj} = 25 \text{ °C}$	490		mJ
			$T_{vj} = 125 \text{ °C}$	630		
			$T_{vj} = 150 \text{ °C}$	665		
Thermal resistance, junction to case	R_{thJC}	per IGBT			17.7	K/kW
Thermal resistance, case to heat sink	R_{thCH}	per IGBT, $\lambda_{grease} = 1 \text{ W}/(\text{m} \cdot \text{K})$		10.4		K/kW
Temperature under switching conditions	$T_{vj op}$		-40		150	°C

5 Diode, 3-Level

Table 9 **Maximum rated values**

Parameter	Symbol	Note or test condition	Values	Unit
Repetitive peak reverse voltage	V_{RRM}	$T_{vj} = 25 \text{ °C}$	2300	V
Continuous DC forward current	I_F		1800	A

(table continues...)

Table 9 (continued) Maximum rated values

Parameter	Symbol	Note or test condition	Values	Unit
Repetitive peak forward current	I_{FRM}	$t_p = 1 \text{ ms}$	3600	A

Table 10 Characteristic values

Parameter	Symbol	Note or test condition	Values			Unit
			Min.	Typ.	Max.	
Forward voltage	V_F	$I_F = 1800 \text{ A}$, $V_{GE} = 0 \text{ V}$	$T_{vj} = 25 \text{ °C}$	3.25	3.64	V
			$T_{vj} = 125 \text{ °C}$	3.00	3.33	
			$T_{vj} = 150 \text{ °C}$	2.95	3.22	
Peak reverse recovery current	I_{RM}	$V_{CC} = 600 \text{ V}$, $I_F = 1800 \text{ A}$, $V_{GE} = -15 \text{ V}$, $-di_F/dt = 8200 \text{ A}/\mu\text{s}$ ($T_{vj} = 150 \text{ °C}$)	$T_{vj} = 25 \text{ °C}$	1120		A
			$T_{vj} = 125 \text{ °C}$	1450		
			$T_{vj} = 150 \text{ °C}$	1530		
Recovered charge	Q_r	$V_{CC} = 600 \text{ V}$, $I_F = 1800 \text{ A}$, $V_{GE} = -15 \text{ V}$, $-di_F/dt = 8200 \text{ A}/\mu\text{s}$ ($T_{vj} = 150 \text{ °C}$)	$T_{vj} = 25 \text{ °C}$	295		μC
			$T_{vj} = 125 \text{ °C}$	580		
			$T_{vj} = 150 \text{ °C}$	665		
Reverse recovery energy	E_{rec}	$V_{CC} = 600 \text{ V}$, $I_F = 1800 \text{ A}$, $V_{GE} = -15 \text{ V}$, $-di_F/dt = 8200 \text{ A}/\mu\text{s}$ ($T_{vj} = 150 \text{ °C}$)	$T_{vj} = 25 \text{ °C}$	170		mJ
			$T_{vj} = 125 \text{ °C}$	320		
			$T_{vj} = 150 \text{ °C}$	365		
Thermal resistance, junction to case	R_{thJC}	per diode			39.3	K/kW
Thermal resistance, case to heat sink	R_{thCH}	per diode, $\lambda_{grease} = 1 \text{ W}/(\text{m}\cdot\text{K})$		15.4		K/kW
Temperature under switching conditions	$T_{vj\text{ op}}$		-40		150	°C

Note: Dynamic data for 3-level valid in conjunction with FF2400RB12IP7.
 $T_{vj\text{ op}}$ up to 175 °C is allowed for operations in overload conditions. For detailed specifications please refer to AN2021-11.

6 NTC-Thermistor

Table 11 Characteristic values

Parameter	Symbol	Note or test condition	Values			Unit
			Min.	Typ.	Max.	
Rated resistance	R_{25}	$T_{NTC} = 25 \text{ °C}$		5		kΩ
Deviation of R_{100}	$\Delta R/R$	$T_{NTC} = 100 \text{ °C}$, $R_{100} = 493 \text{ Ω}$	-5		5	%

(table continues...)

Table 11 (continued) Characteristic values

Parameter	Symbol	Note or test condition	Values			Unit
			Min.	Typ.	Max.	
Power dissipation	P_{25}	$T_{NTC} = 25\text{ °C}$			20	mW
B-value	$B_{25/50}$	$R_2 = R_{25} \exp[B_{25/50}(1/T_2 - 1/(298,15\text{ K}))]$		3375		K
B-value	$B_{25/80}$	$R_2 = R_{25} \exp[B_{25/80}(1/T_2 - 1/(298,15\text{ K}))]$		3411		K
B-value	$B_{25/100}$	$R_2 = R_{25} \exp[B_{25/100}(1/T_2 - 1/(298,15\text{ K}))]$		3433		K

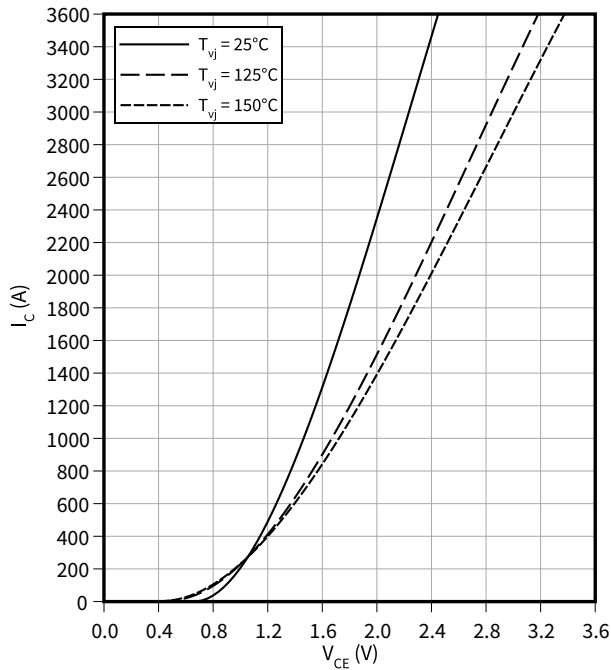
Note: For an analytical description of the NTC characteristics please refer to AN2009-10, chapter 4.

7 Characteristics diagrams

Output characteristic (typical), IGBT, Inverter

$I_C = f(V_{CE})$

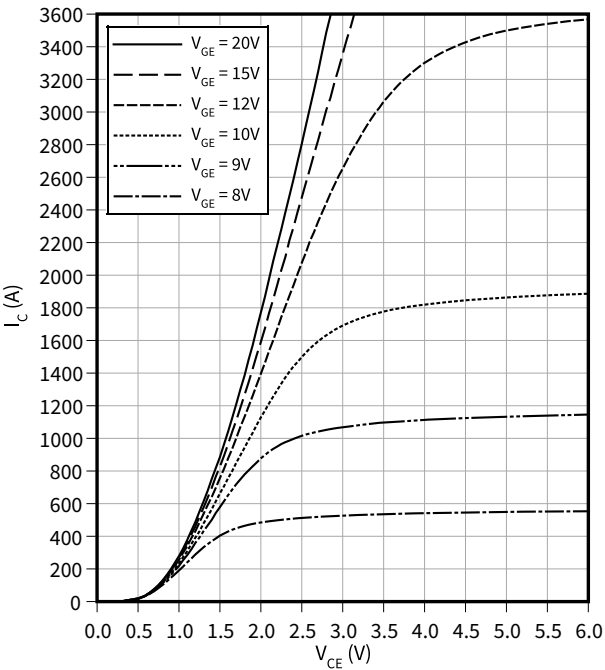
$V_{GE} = 15\text{ V}$



Output characteristic field (typical), IGBT, Inverter

$I_C = f(V_{CE})$

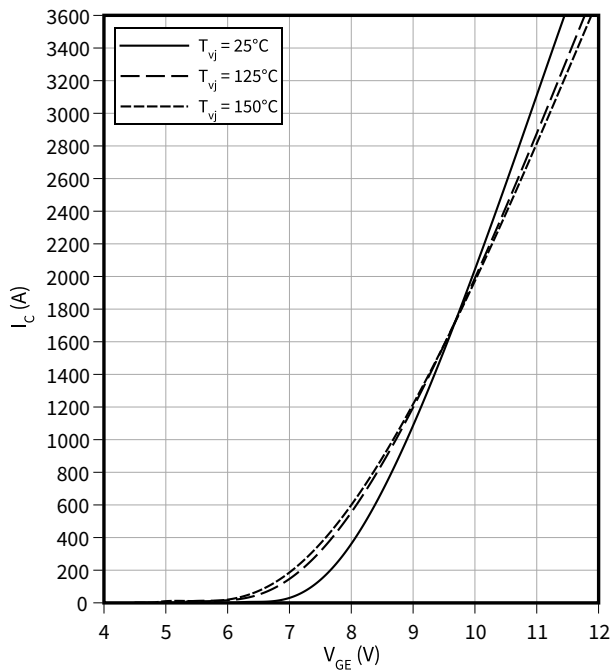
$T_{vj} = 150\text{ °C}$



Transfer characteristic (typical), IGBT, Inverter

$I_C = f(V_{GE})$

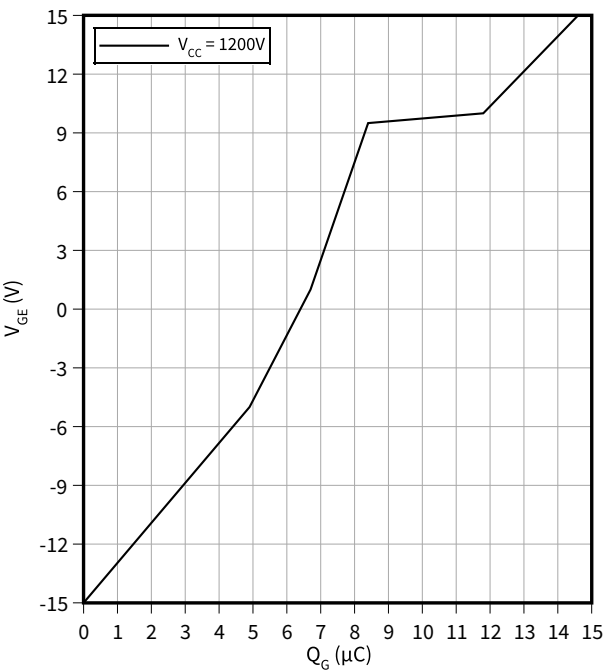
$V_{CE} = 20\text{ V}$



Gate charge characteristic (typical), IGBT, Inverter

$V_{GE} = f(Q_G)$

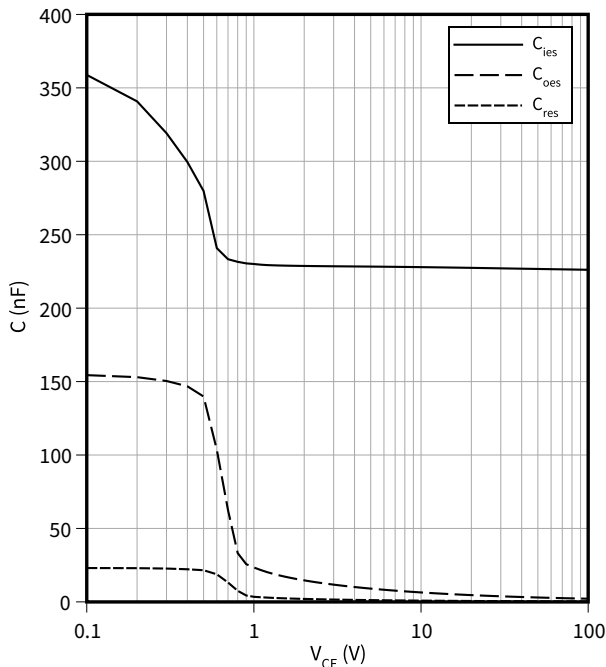
$I_C = 1800\text{ A}, T_{vj} = 25\text{ °C}$



7 Characteristics diagrams

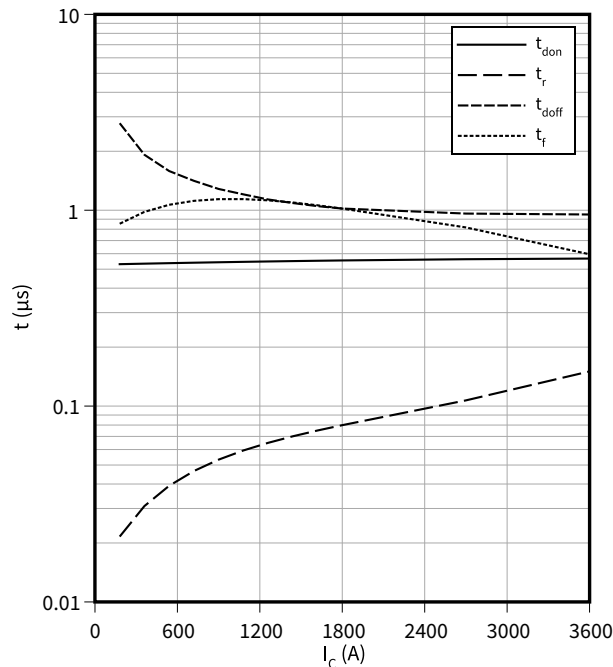
Capacity characteristic (typical), IGBT, Inverter

$C = f(V_{CE})$
 $f = 100\text{ kHz}$, $V_{GE} = 0\text{ V}$, $T_{vj} = 25\text{ °C}$



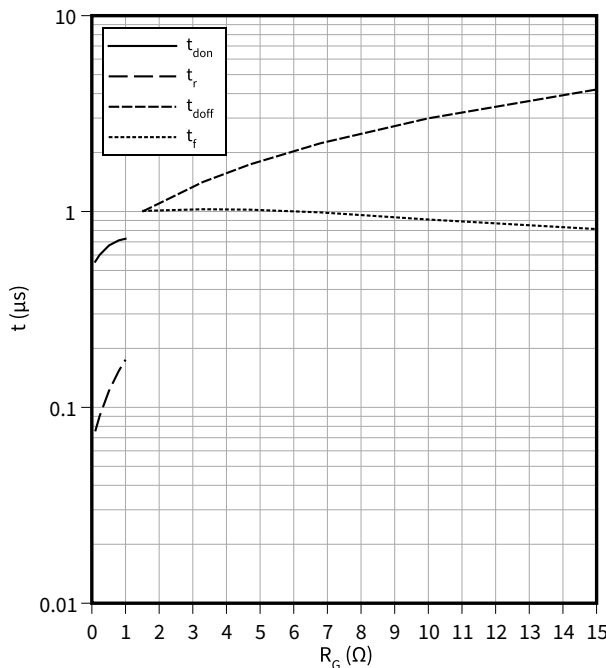
Switching times (typical), IGBT, Inverter

$t = f(I_C)$
 $R_{Goff} = 1.5\text{ }\Omega$, $R_{Gon} = 0.1\text{ }\Omega$, $V_{GE} = \pm 15\text{ V}$, $V_{CC} = 1200\text{ V}$,
 $T_{vj} = 150\text{ °C}$



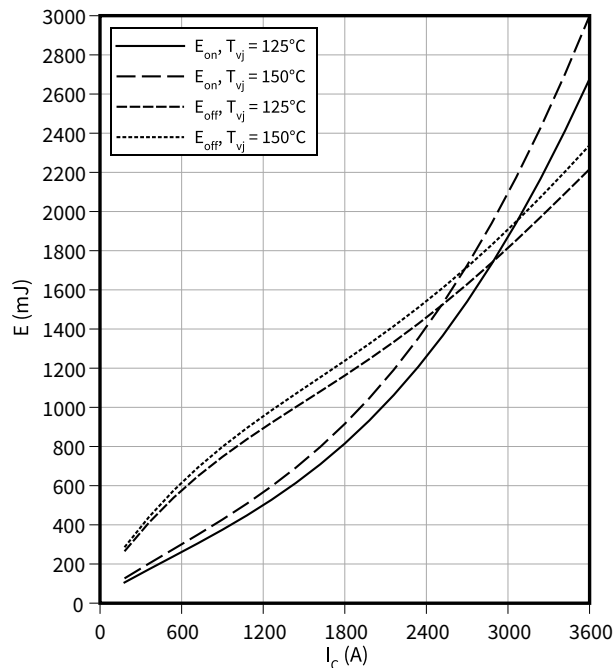
Switching times (typical), IGBT, Inverter

$t = f(R_G)$
 $V_{GE} = \pm 15\text{ V}$, $I_C = 1800\text{ A}$, $V_{CC} = 1200\text{ V}$, $T_{vj} = 150\text{ °C}$



Switching losses (typical), IGBT, Inverter

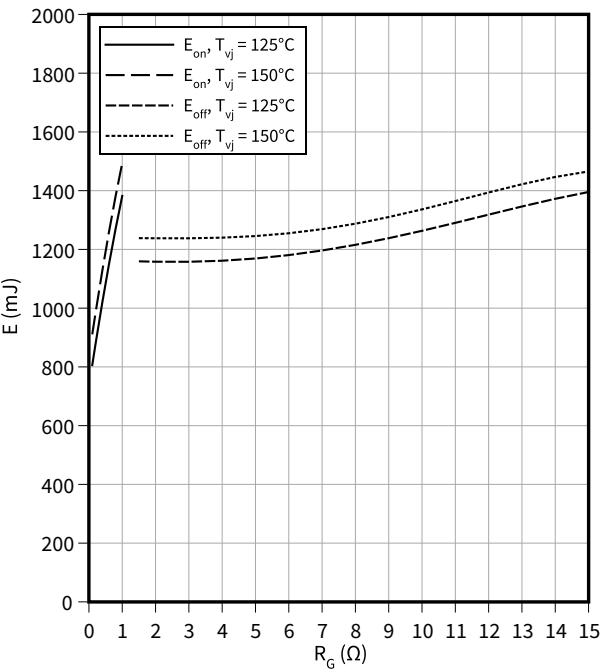
$E = f(I_C)$
 $R_{Goff} = 1.5\text{ }\Omega$, $R_{Gon} = 0.1\text{ }\Omega$, $V_{GE} = \pm 15\text{ V}$, $V_{CC} = 1200\text{ V}$



7 Characteristics diagrams

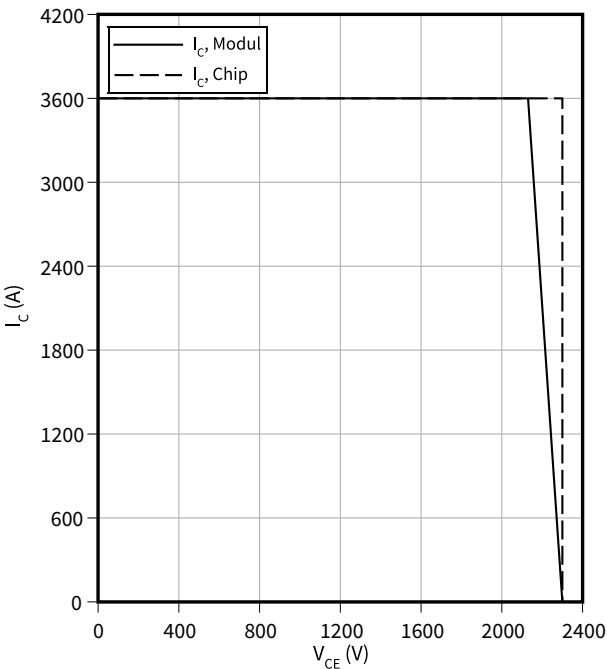
Switching losses (typical), IGBT, Inverter

$E = f(R_G)$
 $V_{GE} = \pm 15\text{ V}$, $I_C = 1800\text{ A}$, $V_{CC} = 1200\text{ V}$



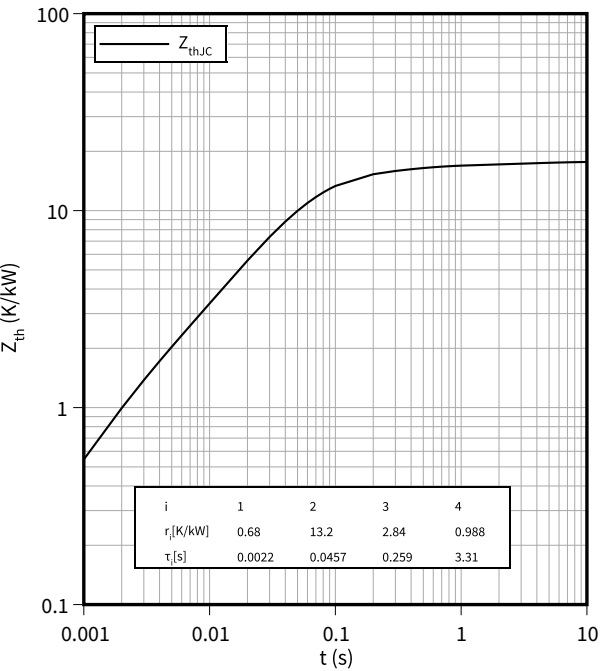
Reverse bias safe operating area (RBSOA), IGBT, Inverter

$I_C = f(V_{CE})$
 $R_{Goff} = 1.5\text{ }\Omega$, $V_{GE} = \pm 15\text{ V}$, $T_{vj} = 150\text{ }^\circ\text{C}$



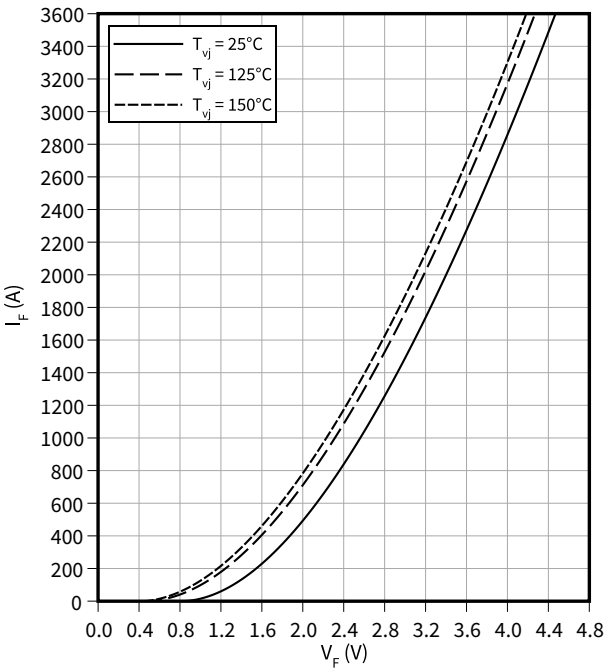
Transient thermal impedance, IGBT, Inverter

$Z_{th} = f(t)$



Forward characteristic (typical), Diode, Inverter

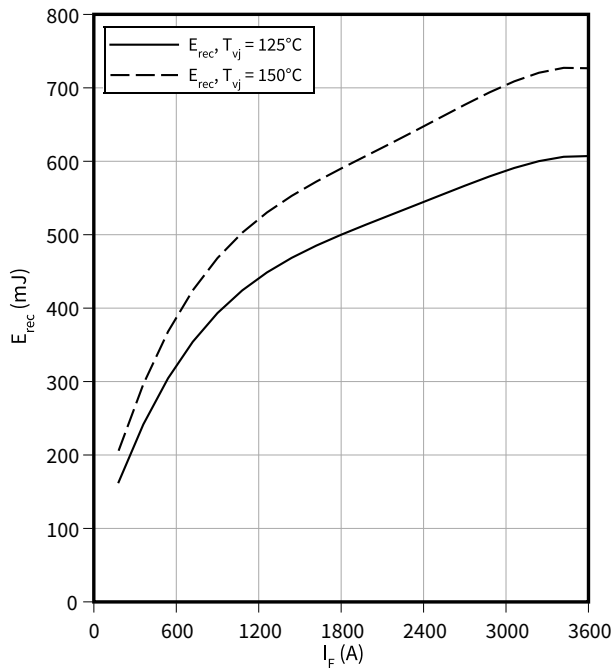
$I_F = f(V_F)$



7 Characteristics diagrams

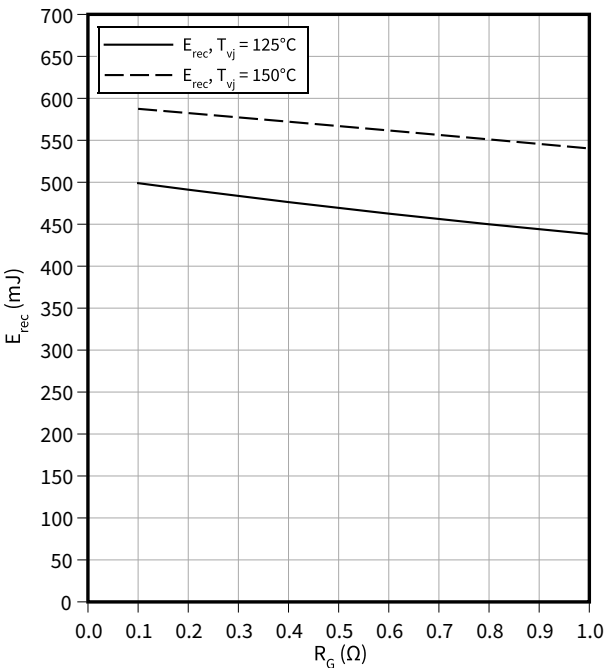
Switching losses (typical), Diode, Inverter

$E_{rec} = f(I_F)$
 $V_{CE} = 1200\text{ V}, R_{Gon} = R_{Gon}(IGBT)$



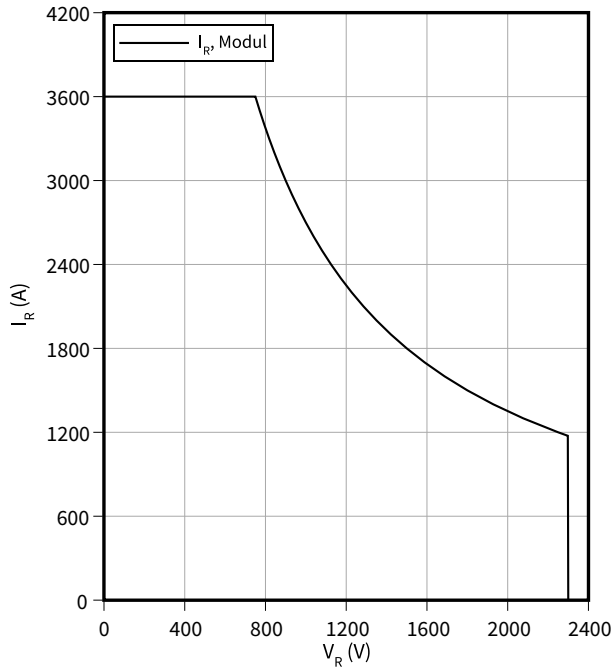
Switching losses (typical), Diode, Inverter

$E_{rec} = f(R_G)$
 $V_{CE} = 1200\text{ V}, I_F = 1800\text{ A}$



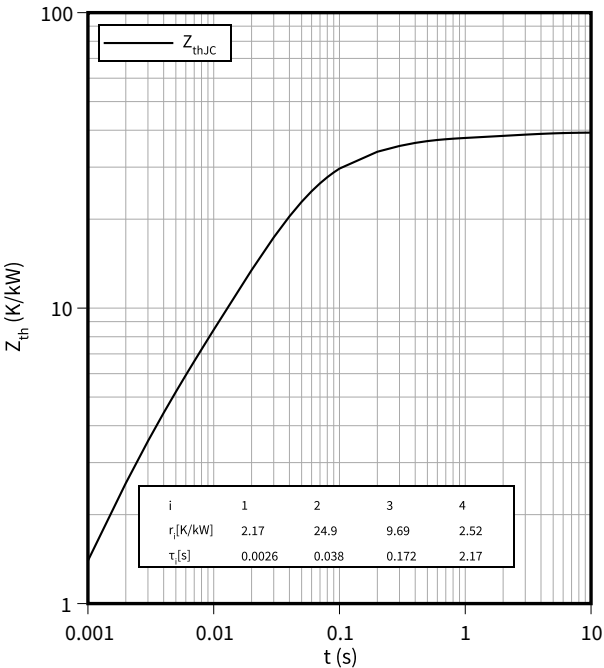
Safe operating area (SOA), Diode, Inverter

$I_R = f(V_R)$
 $T_{vj} = 150\text{ °C}$



Transient thermal impedance, Diode, Inverter

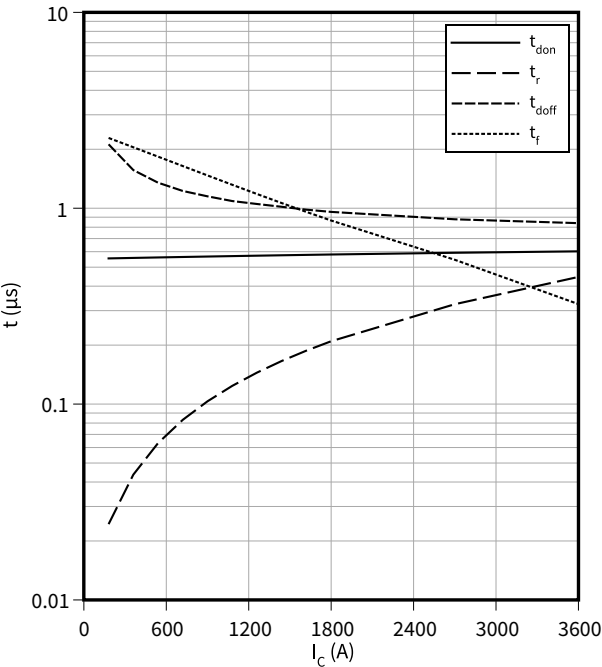
$Z_{th} = f(t)$



7 Characteristics diagrams

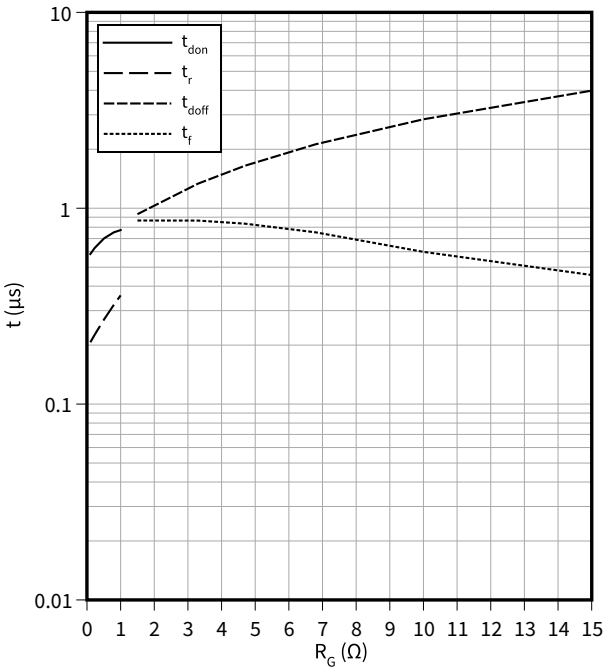
Switching times (typical), IGBT, 3-Level

$t = f(I_C)$
 $R_{Goff} = 1.5 \, \Omega$, $R_{Gon} = 0.1 \, \Omega$, $V_{GE} = \pm 15 \, V$, $V_{CC} = 600 \, V$,
 $T_{vj} = 150 \, ^\circ C$



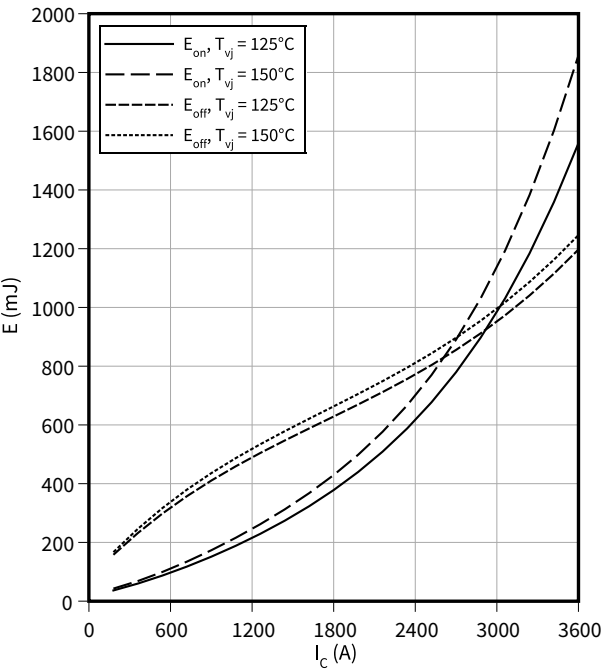
Switching times (typical), IGBT, 3-Level

$t = f(R_G)$
 $V_{GE} = \pm 15 \, V$, $I_C = 1800 \, A$, $V_{CC} = 600 \, V$, $T_{vj} = 150 \, ^\circ C$



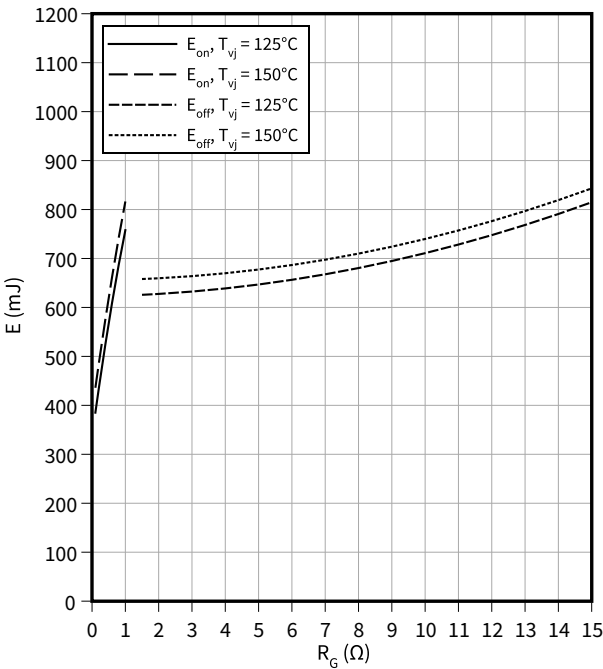
Switching losses (typical), IGBT, 3-Level

$E = f(I_C)$
 $R_{Goff} = 1.5 \, \Omega$, $R_{Gon} = 0.1 \, \Omega$, $V_{GE} = \pm 15 \, V$, $V_{CC} = 600 \, V$



Switching losses (typical), IGBT, 3-Level

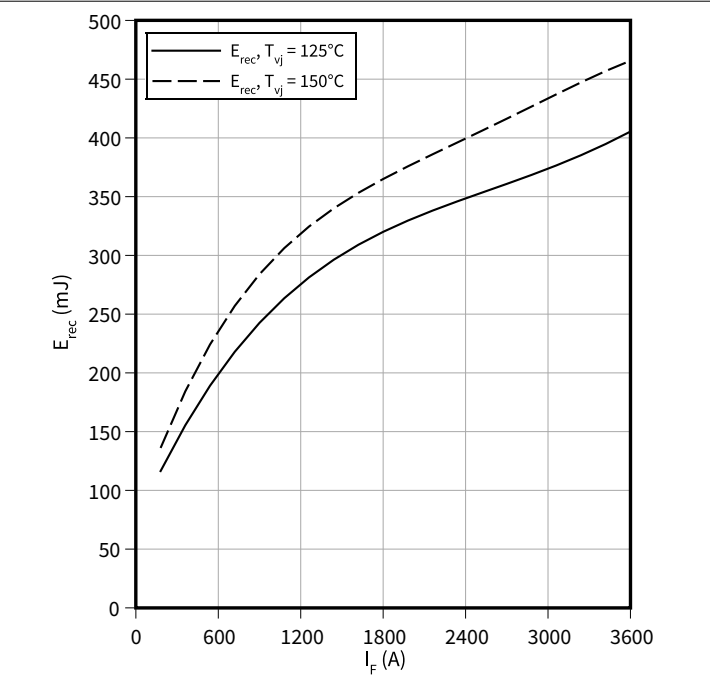
$E = f(R_G)$
 $V_{GE} = \pm 15 \, V$, $I_C = 1800 \, A$, $V_{CC} = 600 \, V$



7 Characteristics diagrams

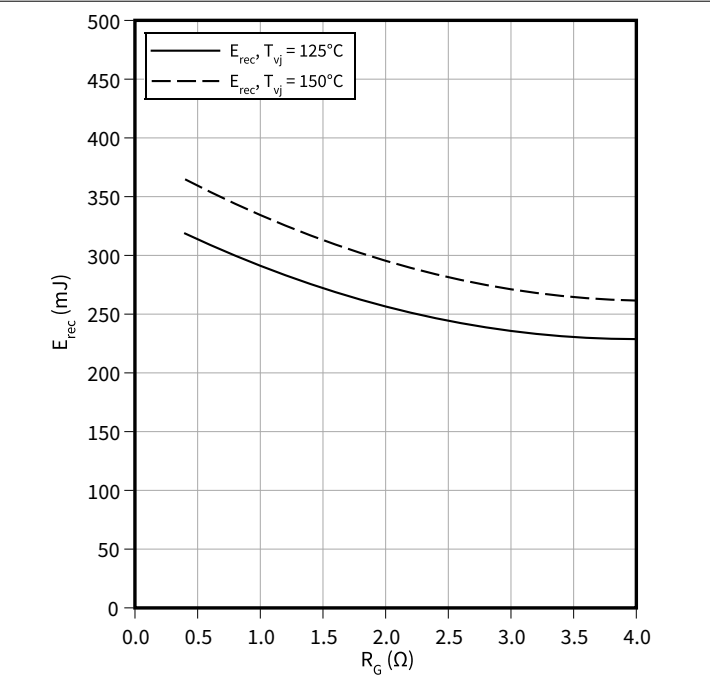
Switching losses (typical), Diode, 3-Level

$E_{rec} = f(I_F)$
 $V_{CE} = 600\text{ V}, R_{Gon} = R_{Gon}(IGBT)$



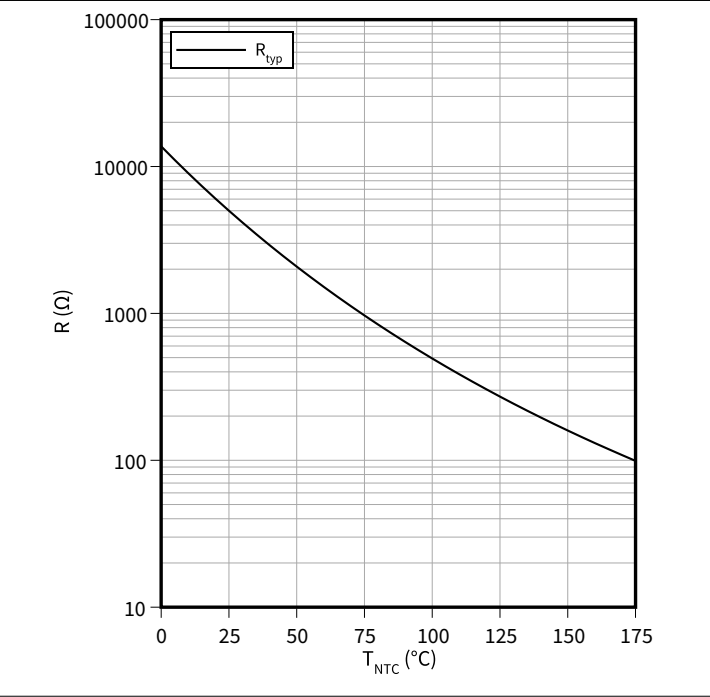
Switching losses (typical), Diode, 3-Level

$E_{rec} = f(R_G)$
 $V_{CE} = 600\text{ V}, I_F = 1800\text{ A}$



Temperature characteristic (typical), NTC-Thermistor

$R = f(T_{NTC})$



8 Circuit diagram

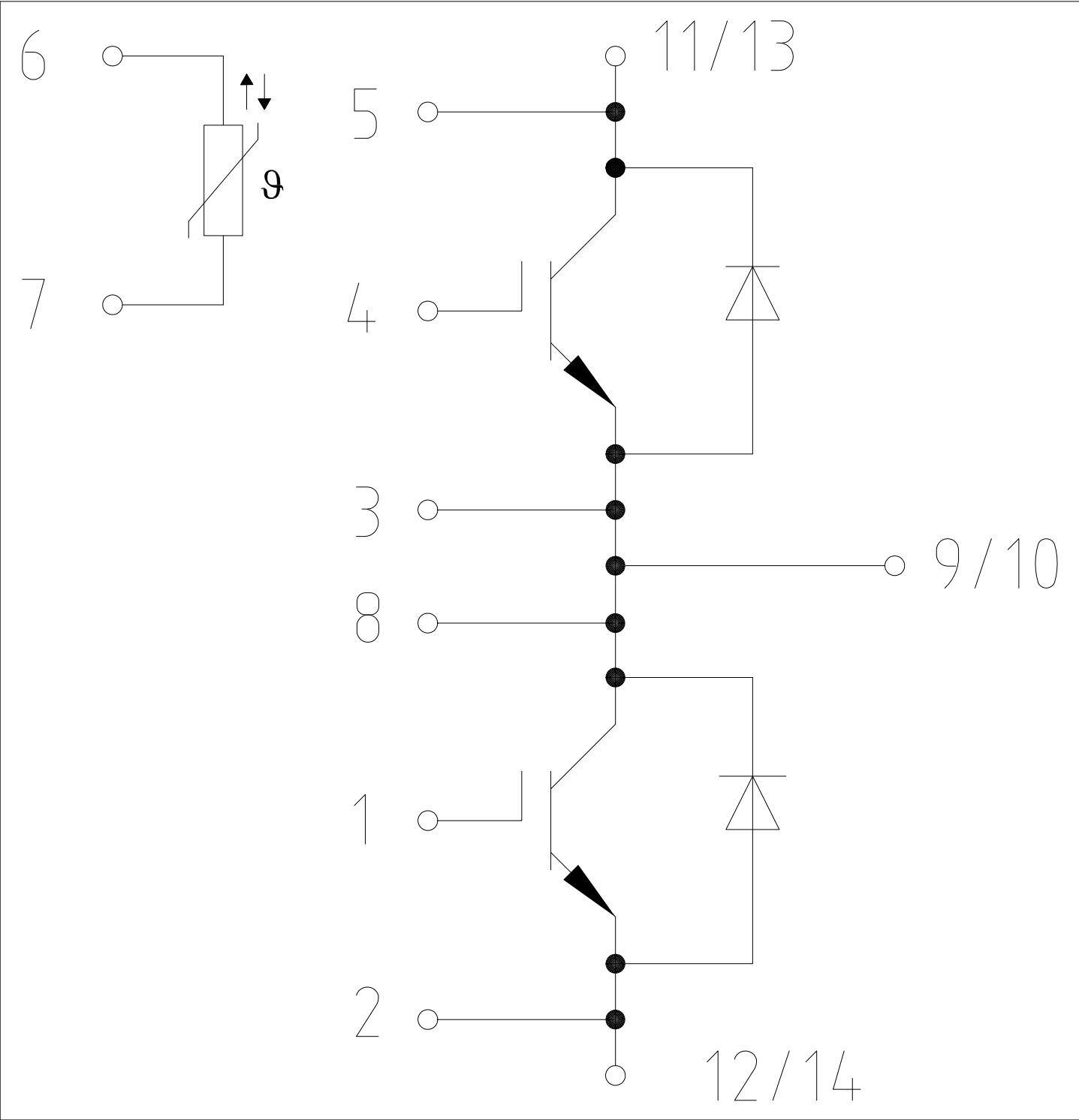


Figure 1

9 Package outlines

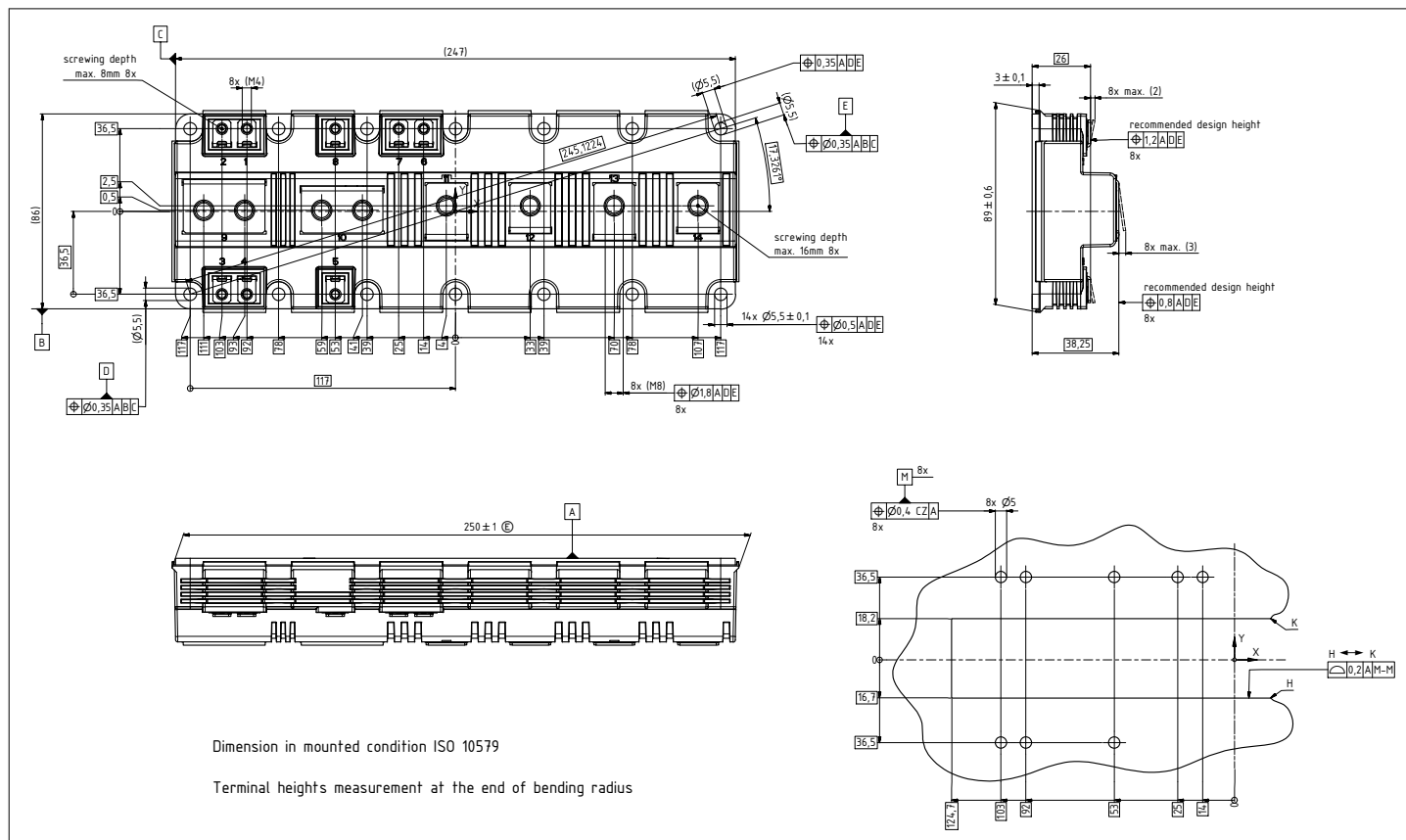


Figure 2

10 Module label code

Module label code			
Code format	Data Matrix		Barcode Code128
Encoding	ASCII text		Code Set A
Symbol size	16x16		23 digits
Standard	IEC24720 and IEC16022		IEC8859-1
Code content	<i>Content</i>	<i>Digit</i>	<i>Example</i>
	Module serial number	1 – 5	71549
	Module material number	6 - 11	142846
	Production order number	12 - 19	55054991
	Date code (production year)	20 – 21	15
	Date code (production week)	22 – 23	30
Example			
			
71549142846550549911530		71549142846550549911530	

Figure 3

Revision history

Document revision	Date of release	Description of changes
V1.0	2019-07-23	Target datasheet
V1.1	2019-12-18	Target datasheet
V1.2	2020-02-14	Target datasheet
n/a	2020-09-01	Datasheet migrated to a new system with a new layout and new revision number schema: target or preliminary datasheet = 0.xy; final datasheet = 1.xy
0.10	2020-12-02	
0.20	2021-02-24	Target datasheet
0.30	2021-08-02	Preliminary datasheet
1.00	2021-08-30	Final datasheet
1.10	2022-04-05	Final datasheet - addition of ton_R
1.20	2022-05-03	Final datasheet - addition of I _{CN}
1.30	2023-07-13	Final datasheet - Correction/ addition of capacity characteristics; Updates of commentary
1.40	2026-06-08	Harmonization of insulation coordination information

Trademarks

All referenced product or service names and trademarks are the property of their respective owners.

Edition 2026-06-08

Published by

Infineon Technologies AG
81726 Munich, Germany

© 2026 Infineon Technologies AG
All Rights Reserved.

Do you have a question about any aspect of this document?

Email: erratum@infineon.com

Document reference
IFX-AA190-011

Important notice

The information given in this document shall in no event be regarded as a guarantee of conditions or characteristics ("Beschaffenhheitsgarantie").

With respect to any examples, hints or any typical values stated herein and/or any information regarding the application of the product, Infineon Technologies hereby disclaims any and all warranties and liabilities of any kind, including without limitation warranties of non-infringement of intellectual property rights of any third party.

In addition, any information given in this document is subject to customer's compliance with its obligations stated in this document and any applicable legal requirements, norms and standards concerning customer's products and any use of the product of Infineon Technologies in customer's applications.

The data contained in this document is exclusively intended for technically trained staff. It is the responsibility of customer's technical departments to evaluate the suitability of the product for the intended application and the completeness of the product information given in this document with respect to such application.

Please note that this product is not qualified according to the AEC Q100 or AEC Q101 documents of the Automotive Electronics Council.

Warnings

Due to technical requirements products may contain dangerous substances. For information on the types in question please contact your nearest Infineon Technologies office.

Except as otherwise explicitly approved by Infineon Technologies in a written document signed by authorized representatives of Infineon Technologies, Infineon Technologies' products may not be used in any applications where a failure of the product or any consequences of the use thereof can reasonably be expected to result in personal injury.